

Gender, Intelligence Theory, and Overplacement:

A Reanalysis of Ehrlinger et al.'s Study 3

1 Introduction

Why do so many people walk out of an exam convinced they did better than they actually did? This is the central question in Ehrlinger et al. (2016), a study connecting beliefs about intelligence to the tendency to overestimate how well we perform relative to our peers.

The phenomenon under study is called *overplacement*: the tendency to rank oneself higher than one actually performs in comparison to others. It has real consequences — students who overplace may not seek help they need, and anyone who misjudges their relative standing is less equipped to decide where to invest effort. Ehrlinger et al. argue that overplacement is shaped by how people think about intelligence. People who hold an *entity theory* — the belief that intelligence is fixed and you either have it or you don't — are especially prone to overplacement, because they selectively focus on easier test questions and draw an inflated sense of how they did overall. Redirecting attention toward harder questions should disrupt this bias. Study 3 tests this idea by randomly assigning participants to review either easy or hard questions after taking a short exam.

Buried in a footnote, the original paper dismisses gender as irrelevant to these patterns. This paper revisits that claim. In the sample used to build the statistical models, men overplaced by roughly 22 percentile points more than women — a difference larger than any other predictor in the analysis. When those models are then tested on a separate group of new participants, however,

the gender advantage reverses. Taken together, the results suggest gender is a real signal within this dataset, but the evidence for it generalizing to new participants is, at best, mixed.

2 Methods

2.1 Participants and Study Design

The data come from Study 3 of Ehrlinger et al. (2016), a controlled lab experiment with 70 undergraduate participants — 39 women and 31 men — recruited from a psychology subject pool. All variables used in this analysis were fully observed; no participants were excluded.

Before the main task, each participant completed a questionnaire measuring their *theory of intelligence*: a continuous score from 1 (entity theory — intelligence is fixed) to 6 (incremental theory — intelligence grows with effort). Participants then took a short exam modeled on the Graduate Record Examination, a standardized test commonly used for graduate school admissions. Afterward, they were randomly assigned to one of two review conditions: the *easy-problems condition* ($n = 27$), in which participants revisited questions most people had answered correctly, or the *hard-problems condition* ($n = 43$), in which they revisited questions most people had found difficult. Each participant then estimated their percentile rank relative to others in the room, and their actual percentile was computed from the group’s real results.

2.2 Measuring Overplacement

Overplacement is the signed difference between a participant’s estimated and actual percentile rank:

$$\text{Overplacement}_i = \text{Estimated Percentile}_i - \text{Actual Percentile}_i. \quad (1)$$

A positive value means the participant thought they ranked higher than they actually did — the hallmark of overplacement. This operationalization follows Moore and Healy (2008) and directly quantifies the gap between self-perception and reality in familiar, interpretable units.

2.3 Statistical Models and Model Comparison

To understand what predicts overplacement, I use ordinary least squares (OLS) regression, a technique that estimates how much each predictor is associated with overplacement on average, while holding the other predictors constant. I compare two models. **Model 1** includes only the theory of intelligence score and the review condition:

$$\text{Overplacement}_i = \beta_0 + \beta_1 (\text{Theory Score})_i + \beta_2 (\text{Hard Condition})_i + \varepsilon_i.$$

Model 2 adds gender as a predictor:

$$\text{Overplacement}_i = \beta_0 + \beta_1 (\text{Theory Score})_i + \beta_2 (\text{Hard Condition})_i + \beta_3 (\text{Male})_i + \varepsilon_i.$$

The decision to add gender is guided by a *partial F-test*, a statistical test that asks whether the improvement in fit from adding gender is larger than what chance variation alone would produce. The null hypothesis is that gender contributes nothing beyond the two predictors already in Model 1.

A key limitation of comparing models using in-sample fit statistics — such as R^2 , the proportion of variance in overplacement explained by the model, or mean squared error (MSE), the average squared prediction error — is that they improve automatically whenever a predictor is added, whether or not that predictor captures real signal. To obtain a fair comparison, I apply both models — fit entirely on the original 70 participants — to a separate *holdout sample*: a second group of participants who completed the same study but played no role in model fitting. Out-of-sample MSE on these new participants is an honest test of whether gender genuinely helps predict overplacement. As a complementary check, I also compute leave-one-out cross-validation (LOOCV) error on the full combined dataset, which works by temporarily removing one participant at a time, refitting the model on everyone else, and predicting the left-out observation.

3 Results

3.1 Exploratory Patterns

Overplacement in the original sample ranged from -62 to $+67$ percentile points, with a mean of 12.2 ($SD = 28.9$). On average, participants believed they ranked about 12 points higher than they actually did. Before fitting any models, two features of the data stand out. First, men and women were essentially equivalent on the other predictors: their theory of intelligence scores were nearly identical (both averaged 3.5 ; $p = 0.87$) and condition assignment was balanced across gender ($p = 0.64$). Second, men overplaced by roughly 22 percentile points more than women on average ($p = 0.001$), a gap that cannot be attributed to differences in beliefs about intelligence or condition assignment. Figure 1 displays these patterns.

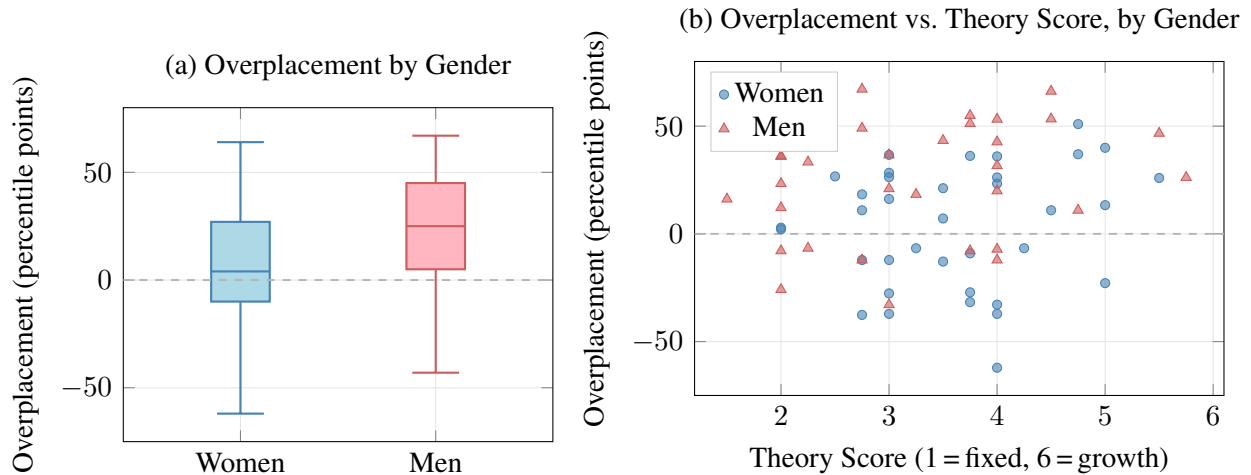


Figure 1: **Exploratory patterns in overplacement.** (a) The typical man (median ≈ 25 pts, red) overplaces substantially more than the typical woman (median ≈ 4 pts, blue). The dashed line marks zero overplacement. (b) Men’s scores (triangles) are shifted upward relative to women’s (circles) across the full range of theory scores, consistent with a gap in overall level rather than a difference in how the two groups respond to the theory predictor.

3.2 Model 1: Theory and Condition Alone

Model 1 explains very little of the variation in overplacement ($R^2 = 0.025$, in-sample $MSE = 800.3$). The estimated coefficient on theory score is slightly positive ($\hat{\beta}_1 = 4.65$), and the hard-

review condition is associated with modestly less overplacement ($\hat{\beta}_2 = -2.09$), a direction consistent with the study’s prediction. With only 70 participants, however, neither effect is estimated precisely enough to rule out chance variation. Overall, knowing a participant’s theory score and review condition tells us very little about how much they will overplace.

Despite fitting poorly overall, Model 1’s residuals — the gaps between predicted and actual overplacement — reveal a striking pattern. Men’s residuals average +12.3 percentile points while women’s average −9.8, a gap of roughly 22 points that the model cannot account for ($t = -3.46$, $p = 0.001$). Figure 2 displays these residuals. The trend lines in panel (b) run roughly parallel across theory scores, suggesting the theory-score relationship is similar for men and women and that a simple shift in overall level — not an interaction — is what gender adds.

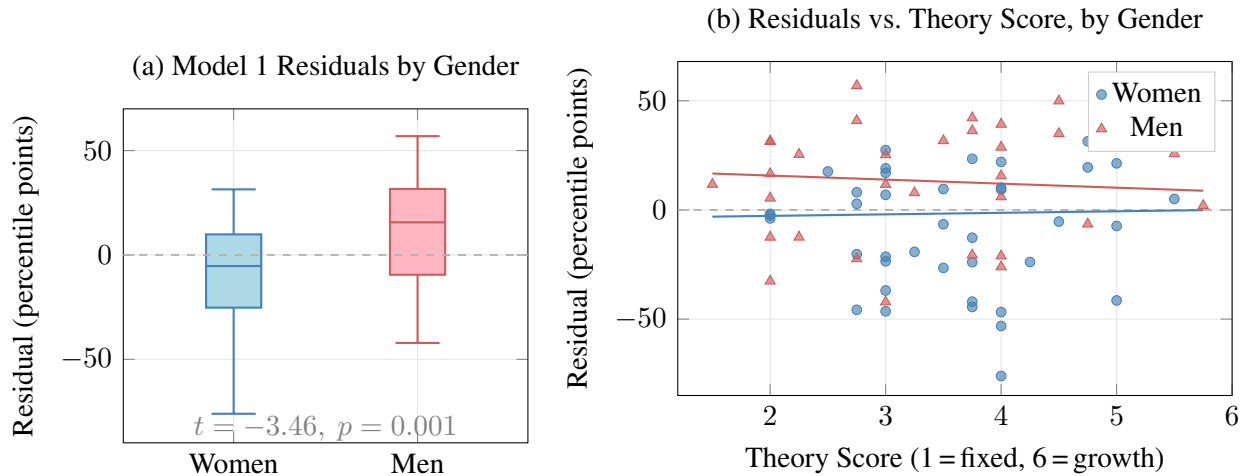


Figure 2: **Model 1 residuals reveal a gender gap.** A residual is the difference between a participant’s actual overplacement and what Model 1 predicted; a positive value means the model underestimated. **(a)** Men’s residuals (red) sit mostly above zero while women’s (blue) sit mostly below, a gap of about 22 points ($t = -3.46$, $p = 0.001$). **(b)** The trend lines run roughly parallel across theory scores, indicating the gap is a consistent level shift rather than a difference in how theory relates to overplacement.

3.3 Model 2: Adding Gender

Adding gender as a main effect produces a clear improvement over Model 1 on the data used for fitting ($F(1, 66) = 11.9$, $p = 0.001$): R^2 rises from 0.025 to 0.174 and in-sample MSE drops from

800.3 to 678.1. The model estimates that, after accounting for theory score and review condition, men overplace by 22.5 percentile points more than women on average ($SE = 6.5$). Figure 3 shows the fitted lines from Model 2. All four lines run parallel, confirming the absence of meaningful gender interactions: the male lines simply sit about 22 points above the female lines regardless of theory score or condition.

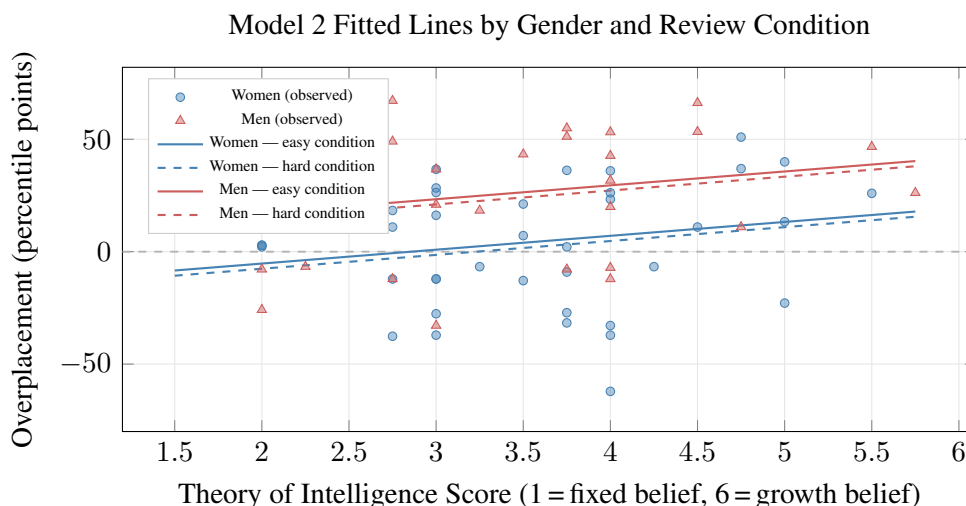


Figure 3: **Model 2 fitted predictions.** Circles are observed values for women; triangles for men. Solid lines show predicted overplacement for the easy-problems condition; dashed lines for the hard-problems condition. All four lines share the same slope. The male lines (red) sit approximately 22 percentile points above the female lines (blue) at every theory score. The dashed lines fall slightly below the solid lines for both genders, consistent with the hard-review condition modestly reducing overplacement.

3.4 Does the Gender Effect Generalize?

The in-sample evidence for gender is compelling, but the more important question is whether a model that knows a participant's gender actually predicts their overplacement better for someone new. To answer this, I applied both models to the holdout sample. The results are surprising.

Model 1, which ignores gender entirely, achieves an out-of-sample MSE of 899.2 on the holdout participants. Model 2, which includes gender, achieves an out-of-sample MSE of 1185.9 — worse by 287 units, a 32% increase in prediction error. A paired t -test comparing the squared prediction error for each holdout participant across the two models confirms this reversal is sta-

tistically meaningful ($t = -2.29$, $p = 0.028$). Rather than helping, gender makes the model's predictions noticeably worse on new data.

Table 1 and Figure 4 summarize the full picture. Leave-one-out cross-validation on the combined dataset ($n = 104$) is slightly more charitable to the gender model (LOOCV MSE of 850.6 vs. 865.5), but the improvement is small. Together, these two out-of-sample evaluations send a mixed signal: gender may carry some genuine predictive value, but the original sample appears too small to estimate the gender coefficient precisely enough that carrying it into new data actually helps.

Table 1: **Model fit statistics.** R^2 is the proportion of variance in overplacement explained. In-sample MSE is computed on the 70 participants used to fit the models; out-of-sample MSE is computed on the separate holdout sample. Lower MSE is better.

Model	Predictors	R^2	In-sample MSE	Out-of-sample MSE
Model 1	Theory + Condition	0.025	800.3	899.2
Model 2	Theory + Condition + Gender	0.174	678.1	1185.9

Note. LOOCV MSE (combined sample, $n = 104$): Model 1 = 865.5; Model 2 = 850.6.

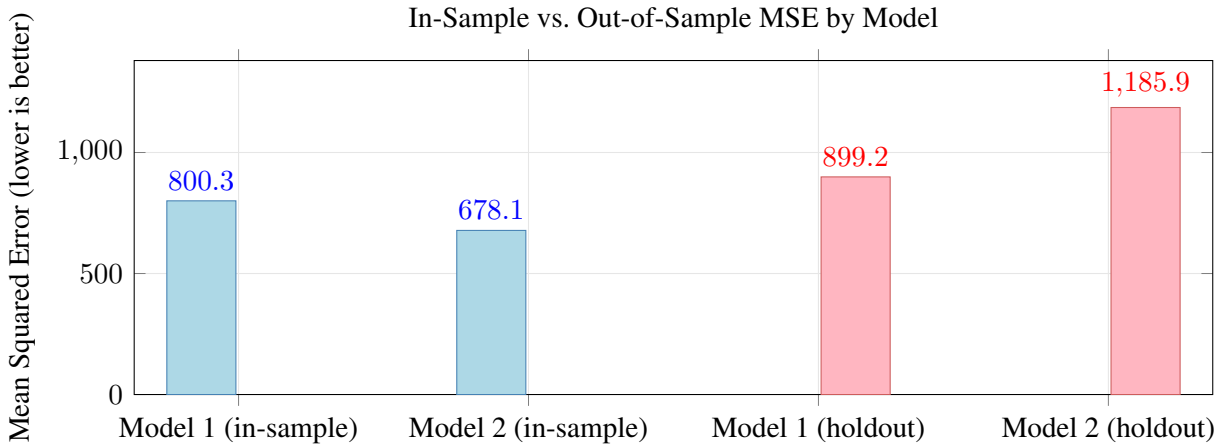


Figure 4: **In-sample fit improves with gender; out-of-sample fit worsens.** Blue bars show MSE on the 70 participants used to fit the models; adding gender reduces in-sample error. Red bars show MSE on the holdout sample; adding gender increases prediction error by 32%, a reversal that is statistically significant ($p = 0.028$). This pattern is the hallmark of a coefficient estimated imprecisely from a small sample.

4 Discussion

This paper set out to ask whether gender plays a meaningful role in overplacement once theory score and review condition are accounted for. The answer is genuinely complicated.

Within the 70 participants used to fit the models, the evidence for gender is strong. Men overplaced by approximately 22 percentile points more than women, a difference that survives adjustment for the other predictors and produces a statistically significant improvement in in-sample fit. On purely within-sample grounds, the original footnote’s dismissal of gender is hard to defend.

The holdout validation, however, introduces serious doubt. When Model 2 is applied to new participants, it predicts their overplacement substantially worse than the simpler Model 1 that ignores gender altogether. The most likely explanation is that 70 participants is simply not enough to estimate a 22-point gender coefficient reliably. A model that learns a larger effect than truly exists in the population will look impressive in-sample but produce miscalibrated predictions elsewhere, which is exactly what is observed here.

What should we conclude? Gender is almost certainly not irrelevant to overplacement. The raw gap is large, the residual pattern is clean, and a broader literature documents consistent gender differences in confidence across academic and financial settings (Beyer, 1990; Barber and Odean, 2001). But this study is not large enough to estimate that effect with enough precision for it to help in new data. The honest summary is: there is enough signal here to motivate a well-powered follow-up study, but not enough to conclude confidently that knowing someone’s gender meaningfully improves overplacement prediction in new samples.

References

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